Open source enterprise stack: Value mapping and use cases

Deploying open source software across the IoT value chain eliminates recurring licence overhead and protects enterprises from platform lock-in. The following matrix outlines the strategic selection of open tools across enterprise layers.

Architecture layer	Open source technology	Strategic enterprise value	Primary business use cases
Edge containerisation	WasmEdge / Wasmtime	Lightweight, cross-platform runtime that isolates edge apps with minimal CPU and memory usage.	Running edge-level inventory reconciliation and sensor logic on low-cost retail gateways.
Edge messaging broker	EMQX / Eclipse Mosquitto	Enterprise-grade MQTT broker supporting millions of concurrent, low-bandwidth sensor connections.	Aggregating real-time telemetry across fleet vehicles, warehouses, and retail storefronts.
Stream processing	Apache Flink	Stateful stream engine capable of processing sub-second complex event processing (CEP) at scale.	Correlating multi-sensor cold-chain readings with active ERP purchase orders in real time.
Event backbone	Apache Kafka / NATS	High-throughput distributed log that acts as the central messaging spine for all enterprise events.	Decoupling physical IoT sensor events from core business systems like ERP, WMS, and CRM.
Open table format	Apache Iceberg / Delta Lake	Provides ACID transactions and time-travel capability on low-cost object storage.	Storing years of supply chain telemetry in open formats for auditability, AI training, and vendor analytics.
Federated analytics	Trino / Apache Spark	High-speed, distributed SQL engine that queries open lakehouse data without proprietary ETL pipelines.	Running cross-functional analytics across historical IoT telemetry, supplier performance, and financial data.

Architectural principles for C-suite leaders

Building an open source IoT architecture requires clear governance to ensure long-term stability, compliance, and ROI.

Own your data standards: Never store operational telemetry in proprietary vendor formats. Mandate open storage standards like Apache Parquet governed by open table formats like Apache Iceberg. This ensures your supply chain data remains fully queryable by modern AI models and analytical engines for years without migration costs.

Decouple edge from cloud: Design field systems to operate autonomously during network disconnects. Gateways must buffer transactions locally using store-and-forward patterns to prevent operational downtime during internet outages.

Focus on event-driven ERP actions: Avoid polling databases. Configure your IoT stream layer to emit business-level events (e.g., ShipmentDelayed, ThresholdBreached, ShelfDepleted) directly into enterprise service buses, triggering immediate automated workflows inside ERP systems.

The enterprise value of IoT does not lie in the hardware—it lies in the velocity and quality of the operational decisions it enables. By deploying a modular, open source stack from the edge to the open lakehouse, forward-thinking organisations eliminate expensive vendor lock-in, streamline supply chain operations, and build a resilient data infrastructure that turns real-time physical events into sustainable competitive advantage. END

References

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- Apache Software Foundation: Apache Iceberg Official Documentation — Official specification and technical guide for open lakehouse table formats and stateful stream processing integrations

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